THE COTANGENT COMPLEX OF A MAP OF SHEAVES OF RINGS IS THE SHEAFIFICATION OF THE SECTIONWISE COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays a commutative square of categories; the reduction to the chaotic topology is the first of its two paragraphs.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- Pullback of the cotangent complex along a morphism of topoi. Lemma lemma-pullback-cotangent-morphism-topoi, used at proof lines 3427-3429. Reduces the claim to the site with the chaotic topology.
- The morphism of topoi given by sheafification and the inclusion of sheaves into presheaves. used at proof lines 3422-3426. Provides the morphism to which the pullback statement is applied.
- Commutation of the sections functor with the relevant categories. used at proof lines 3431-3442. Identifies the value of the complex with the sectionwise complex in the chaotic case.
- The definition of the cotangent complex of a map of sheaves of rings. used at proof lines 3416-3419. Fixes the object whose cohomology is computed.

Source reference. Original lemma 08SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11143–11143. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $\mathcal{C}$ be a site. Let $\mathcal{A} \to \mathcal{B}$ be a homomorphism of sheaves of rings on $\mathcal{C}$. Then $H^i(L_{\mathcal{B}/\mathcal{A}})$ is the sheaf associated to the presheaf $U \mapsto H^i(L_{\mathcal{B}(U)/\mathcal{A}(U)})$.*

Proof. Let $\mathcal{C}'$ be the site we get by endowing $\mathcal{C}$ with the chaotic topology (presheaves are sheaves). There is a morphism of topoi $f : Sh(\mathcal{C}) \to Sh(\mathcal{C}')$ where f_* is the inclusion of sheaves into presheaves and f^{-1} is sheafification. By Lemma 08SV it suffices to prove the result for $\mathcal{C}'$, i.e., in case $\mathcal{C}$ has the chaotic topology.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

If $\mathcal{C}$ carries the chaotic topology, then $L_{\mathcal{B}/\mathcal{A}}(U)$ is equal to $L_{\mathcal{B}(U)/\mathcal{A}(U)}$ because

$$\begin{array}{ccc} \mathcal{A}\text{-}Alg & \rightleftarrows & Sh(\mathcal{C}) \\ \Big\downarrow{\scriptstyle \text{sections over } U} & & \Big\downarrow{\scriptstyle \text{sections over } U} \\ \mathcal{A}(U)\text{-}Alg & \rightleftarrows & Sets \end{array}$$

commutes. $\square$